Benchmarking sequence and structure homology based methods for eukaryotic virus protein annotation

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Abstract

Introduction

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Material and Methods

Dataset selection

Test set

To benchmark different methods for eukaryotic virus protein annotation, we selected 10% of the exemplar species in ICTV's Virus Metadata Resource (VMR) v40.2 that did not have bacteria or archaea as designated host (n=1,000). We subsequently extracted their GenBank accession numbers (n=1,402) and downloaded all associated proteins (n=11,360).

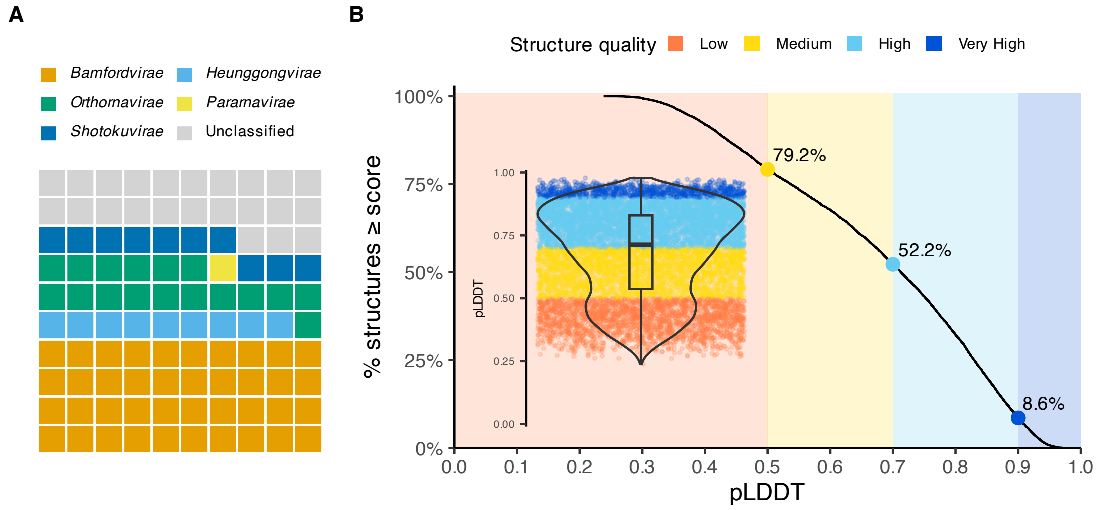

Figure 1: A) Waffle plot showing the distribution of the number of proteins from the different viral Kingdoms. Each tile roughly corresponds to 110 proteins. B) A cumulative distribution function plot showing the percentage of predicted structures from the test set with a pLDDT score above the pLDDT score on the x-axis. The inset shows a violin- and boxplot with all pLDDT scores from the predicted structures (n=11,467).

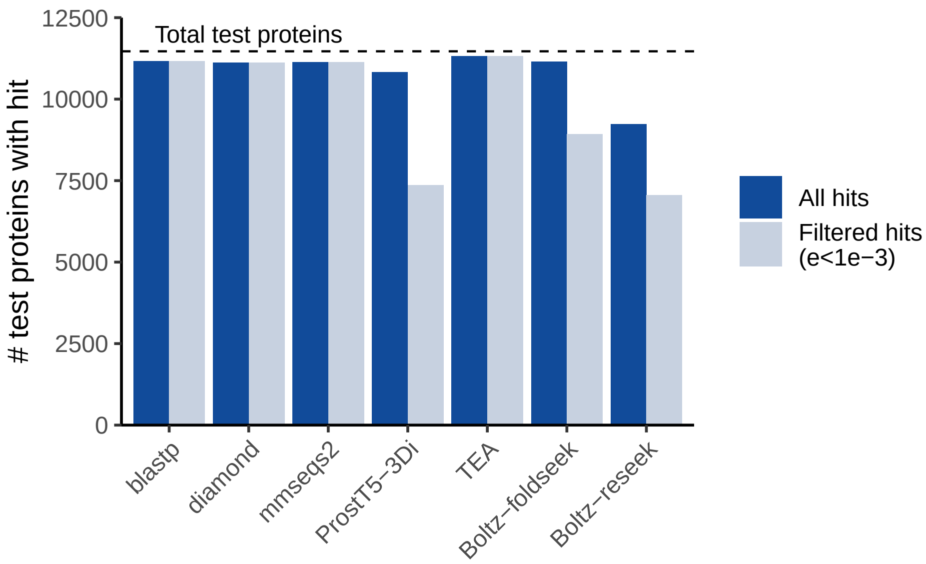

Figure 2: Bar plot showing the number of proteins that got a hit per method, before and after filtering the reported results on e-value < 1e-3.

Protein sequences that were longer than 2,500 amino acids were split into multiple chunks of maximum 2,500 amino acids so that structure prediction with Boltz-2[1] would not run out of GPU memory (see below). This resulted in a final test set of 11,467 protein sequences from 1,000 eukaryotic virus species.

Database

The BFVD[2] was used as the reference database for both sequence and structure homology-based annotation. The BFVD is a comprehensive database of predicted protein structures from the viral sequence representatives of the UniRef30 clusters. UniRef30 is a UniProt database that clusters protein sequences at 30% sequence identity and selects a representative sequence for each cluster, these representatives thus have a UniProt accession. Since a substantial amount of BFVD's included proteins are no longer available in UniProtKB as they are not part of reference proteomes, we converted their UniProt accessions to UniParc IDs.

Functional categories for each BFVD entry were derived from protein names associated with the corresponding UniParc IDs (could be from UniProtKB/TrEMBL, RefSeq and/or GenBank) for eukaryotic viruses (also see below) or by classifying them into PHROG-like[3] categories with Empathi[4] in the case of bacteriophage proteins. The distinction between eukaryotic virus and bacteriophage proteins was made based on the taxid information from the UniParc protein ID.

The BFVD database was further converted into the required formats the search tools used in this benchmarking effort (see sequence and structure homology-based annotation).

Functional classifier

Functional categories for eukaryotic virus proteins were designed with Perplexity AI in deep research mode using the following prompt: "Can you give me an exhaustive list of common protein functions in eukaryotic viruses? Use https://viralzone.expasy.org and https://ictv.global/report/genome as sources. Give me a machine readable file with all these functional categories." The resulting functional categories were further manually curated to ensure that they were mutually exclusive. The final list of functional categories is available in Supplementary Table 1.

A custom script generated by Perplexity AI was used to assign functional categories to BFVD proteins based on all their protein names from UniParc in a JSON file. The script uses pattern matching to assign the most appropriate functional category to each protein. If no keywords match, the protein is assigned to the "Hypothetical Protein" category. The prompt to generate the classifier script was: "Write me a script that classifies the proteins in the uploaded JSON file (as specific as possible) into these functional categories, also if there are proteins in this file that do not fit in one of your proposed categories search new sources and create new functional categories.". Followed up by: "Split the polymerase category in RNA-dependent RNA polymerase, DNA polymerase and reverse transcriptase. In addition split the Capsid/Nucleocapsid/Core Structural into Capsid and Nucleocapsid categories. Split helicase and NTPase.".

The pattern matching rules, functional categories, and classification script were subsequently refined through manual curation to optimize the functional assignment based on protein names.

The same functional classifier script was also used on the protein names from our test set to assign functional categories to the eukaryotic virus proteins downloaded from the ICTV exemplar species.

Sequence homology-based annotation

For sequence homology annotation, we used three state-of-the-art tools widely used in functional annotation pipelines and sequence similarity searches. As gold standard, blastp[5] was used with an e-value cutoff of 1e-3. In addition, we used diamond[6] in sensitive mode and mmseqs2[7] with default settings; both have a default e-value cutoff of 1e-3.

Structure prediction

For the structure prediction of our test set proteins, we relied on Boltz-2[1] (default settings) with multiple sequence alignments (MSAs) generated by a local, adapted ColabFold installation[8] (see https://github.com/gbouras13/colabfoldv). To increase the depth and diversity of the MSAs, we used the default ColabFold sequence databases, supplemented with extra *Orthornavirae* sequences[9] and viral sequences used in the structure prediction for the Phold database[10]. Structures were predicted on an NVIDIA H100 GPU with 80GB of VRAM.

Structure homology-based annotation

ProstT5

TEA

Foldseek

Reseek

Computational benchmarking

- hyperfine

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